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Explainable machine learning in cybersecurity: A survey

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Abstract

Machine learning (ML) techniques are becoming more and more important in cybersecurity, as they can quickly analyse and identify different types of threats from millions of events. In spite of the increasing number of possible applications of machine learning, successful adoption of ML models in cybersecurity still highly relies on the explainability of those models that are used for making predictions. Explanations that support ML model outputs are crucial in cybersecurity-oriented ML applications because people need to get more information from the model than just binary output for analysis. Explainable models help ML developers solve the “trust” problem for security application predictions by validating model behaviours, diagnosing misclassifications and sometimes automatically patching errors in the target models. Therefore, explainable ML for cybersecurity has become a necessary and important research branch. In this survey, we present the topic of explainable ML in cybersecurity through two general types of explanations: (1) ante hoc explanation, and (2) post hoc explanation, with their methodologies. Specifically, we systematically review and categorise the state-of-the-art research, and provide

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comparative studies to help researchers find the optimal solutions to specific cybersecurity problems. We also list open issues in this field to facilitate future studies. This survey will benefit diverse groups of readers from both academia and industries, who want to effectively use ML to solve cybersecurity challenges.

KEYWORDS

cybersecurity, explainable approaches, explanation, machine learning

1 | INTRODUCTION

Machine learning (ML) refers to a series of methods that data processing systems use to make and improve predictions.¹ It is fundamental to recent advances in cybersecurity, and it has been successfully integrated into security-relevant systems, for example, malware detection,^{2,3} intrusion detection^{4,5} and vulnerability discovery,^{6,7} wherein ML processes map input data to predicted labels. Predictors can be linear models or neural networks with a set of weights.⁸ Without the need to explicitly program and output results, ML techniques provide security applications with the capability to automatically learn based on past algorithms.⁹

However, due to the lack of explanation, there are various problems in the security field:

Problem 1: Digital applications suffer from cyberthreats, while the users—operators and designers—do not know where the threat comes from. The decision-making processes of the detection models are generally not interpretable to users.¹⁰

Problem 2: There is a gap between predictions and explanations; an ML model predicts without explanation and attempts to come to a conclusion without explanation. Unless explained, ML merely makes guesses.⁹

Problem 3: In contrast to other application domains, cybersecurity poses particular challenges to the need for explanation methods.¹¹ The users tend to trust security systems when explanations are given and corresponded with predictions rather than the system with only calculated predictions.

Problem 4: A large number of learnt models in security applications contain prejudices and human biases.¹² Data-based decision-making models can inherit these biases, and the subsequent prediction from such models may become unfair or even wrong. Explanations in security applications might demonstrate the underlying causalities in the model.

To meet the needs of explainable ML, more and more intensive research on improving the capability of providing explanations of ML methods. It has been proposed that various explainable methods can help target users understand how they work. These methods decide if one should trust a prediction. We demonstrate “what the explanations in cybersecurity give”: validating learnt model behaviours² for ML developers; identifying classification errors;¹³ and patching errors for the learnt model for safety.¹⁴ Moreover, different explanations between various types of detection classifiers enable security developers to better design the security

system structures. With more attention on their applications in cybersecurity, people have been thinking about questions: (1) which explainable methods are suitable for different security systems and (2) what the requirements do these systems need to meet?

Explanations provide trust in ML methods, that exemplify the importance of explainable ML.^{15,16} In recent years, several directions for explaining security systems have been explored. The cybersecurity experts have started to develop models for understanding ML in cybersecurity, for example, Android malware detection,^{2,17,18} portable document format (PDF) malware detection^{11,14,19} and discovering vulnerabilities in source code.^{6,11} These explainable methods can help trace deep learning's prediction back to the input in security to help understand the decision-making process.

In this survey, we introduce the procedure of how to build an explainable method and two general types of explainable ML methods deployed in cybersecurity: ante hoc explanations and post hoc explanations. Further, we present the explanation approaches under two categories in detail, including the cons and pros of these approaches to help select methods on various real-world data sets. This study paves the way for implementing and trusting robust neural models accompanied by explanations for their predictions.²⁰ Comparative studies are provided based on rigorous evaluation metrics to help researchers find the optimal explainable approaches for their work. Finally, we systematically summarise and analyse open issues with an event timeline. This will help readers understand explanatory developments, milestones, practical implications and critical problems.

The rest of this survey is organised as follows. An overview of two types of explainable processing systems' methodologies is given in Section 2. We systematically review and categorise the technical background of each state-of-the-art explainable learning method in Sections 3 and 4, respectively. Comparative studies are provided according to rigorous evaluation metrics in Section 5. The explanation summary for open issues is described in Section 6. Section 7 presents the discussion for future research, followed by the conclusion in Section 8.

2 | EXPLANATIONS IN CYBERSECURITY

2.1 | Role of explanations in cybersecurity

A number of papers examined different types of security threats and tried to identify the causes,^{7,21,22} but there was no significant reduction in threats based on the results. Explanations play an important role in guiding users through the decision-making process and identifying the causes in the security domain. ML explanations in security systems provide people with a better understanding: (1) what the security system model is; (2) which features of an instance are malicious and why it can be attacked; and (3) how to use a system securely.²³

Explanations for ML have been explored with remarkable success in several application fields, for example, natural language processing (NLP)²⁴ and speech recognition,²⁵ while their applications have not received much attention in cybersecurity so far. Figure 1 shows the aspects where explainable machine learning has been applied in security. Specifically, this survey deals with the following types of applications in the security domain:

1. *Malware detection*: There are two kinds of malware detection systems that employ explanations: (1) Android malware detection. It uses ML models to identify Android malware by building features from application's codes and manifests, such as request permissions and suspicious application programming interface calls. Recently there has been more and more research on Android malware detection systems using deep learning.²⁶ Typical Android malware detection

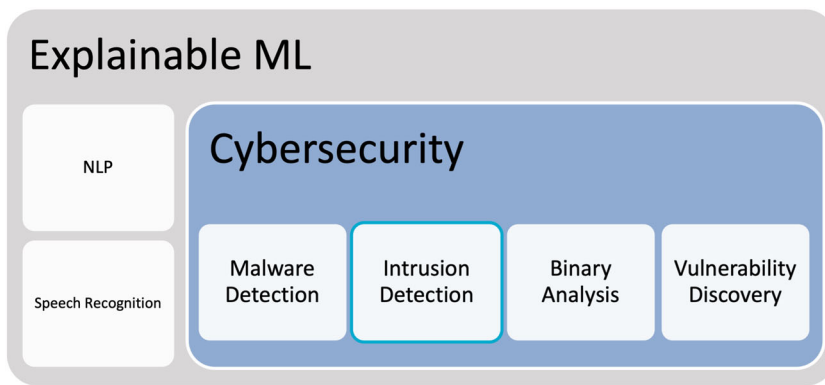


FIGURE 1 Explainable ML for cybersecurity. Four types of applications in the security domain have applied explainable ML. ML, machine learning; NLP, natural language processing. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/int.23088)]

systems include Drebin² and deep android malware detection (DAMD).³ (2) PDF malware detection. The PDF detection systems, for example, the Mimicus+ system, are designed to detect malicious PDF documents. Training is based on PDF document features, including document structure data.¹⁴

2. *Intrusion detection*: The intrusion detection, such as Network Intrusion Detection Systems (NIDSs), enables system operators to detect their systems' network security breaches. The data set is derived from NetFlow traces captured by the real network.¹⁹ One of the NIDSs applies Adversarial Auto Encoder⁴ to do task for Anomaly Detection. The NIDS encounters unpredictable attacks when building a detection model.²⁷
3. *Binary analysis*: Binary reverse engineering reinforces the security of software with a well-trained Recurrent Neural Networks (RNNs) model. Binary analysis helps security analysts through manual work such as detecting the function start.¹⁴
4. *Vulnerability discovery*: Recent attacks make use of diverse vulnerabilities (e.g., cross-domain vulnerability²⁸). There has been some research conducted on the application of ML techniques to assist in the detection of vulnerabilities.²⁹ For example, VulDeePecker is a discovery system using RNN to analyse vulnerabilities in C/C++ code.⁶

The security of these systems refers to the security properties of their stakeholders (e.g., the developer, the client, the analyst and the defender). The security properties include confidence, trust and transparency. All stakeholders want security, which is the primary reason why explainable ML is vital to security systems. Explainable ML is used to explain predictions in these systems, e.g., for individual prediction of a black-box ML model and local boundary of the model in both linear and nonlinear way.

2.2 | Explanation setup in cybersecurity

Explainable methods for ML in cybersecurity can be distinguished by whether the explanations are achieved by a self-explanatory learning model (ante hoc) or by an already trained model (post hoc).

Ante hoc and post hoc explanations in cybersecurity use their own design systems. The workflows of building explanations to users in ante hoc and post hoc are shown in Figures 2 and 3, respectively.

1. *Ante hoc explanation setup*: As shown in ante hoc explanation workflow in Figure 2, the explanatory models incorporate the modules that generate explanations into their architecture to provide explanations for their predictions. These models use different ML processes to train predictions associated with explainable models. The functions of these models have the capacity of predicting and are capable of explaining why it makes that prediction as well. The explainable output displays additional information that can provide insights into why such a prediction was made.

Ante hoc explanations restrict the complexity of the model with the process of explainable modelling. It develops inherently more explainable models. Ante hoc explanations are applied to intrinsically interpretable modelling.

2. *Post hoc explanation setup*: Post hoc explanations mean applying an explainable method to a target model, which is generally a black-box model. Post hoc explanations focus on interpretable methods (e.g., feature importance and partial dependency graph) after training.³⁰ Post hoc explainable methods are used to extract explanations to describe predeveloped models, and can also be treated as proxy models or surrogate models. Post hoc explanations can be deployed to explainable classifications.³¹

As shown in Figure 3, the post hoc explainable methods have a workflow with two interconnected modules: (1) the prediction module is dedicated to making predictions for

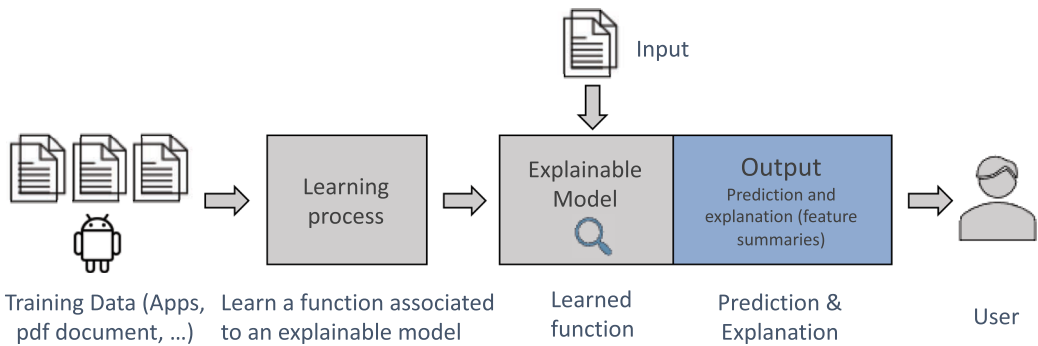


FIGURE 2 Ante hoc explanation workflow in cybersecurity: learned function outputs predictions and explanations for the input [Color figure can be viewed at wileyonlinelibrary.com]

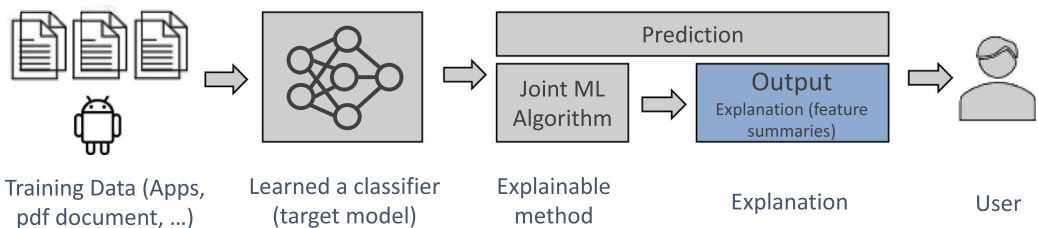


FIGURE 3 Post hoc explanation workflow in cybersecurity: the learned classifier outputs predictions, and the joint ML algorithm acts as an explainable method and outputs explanations results [Color figure can be viewed at wileyonlinelibrary.com]

the task at hand, and (2) the explanation module is dedicated to making interpretations. The process of making an interpretation is a multi-task learning (MTL)³² process, which is able to combine different interpretation algorithms to make an interpretation at the same time. The explanation module does not influence the predictions made by the prediction module.

2.3 | Taxonomy of explanation in cybersecurity

The taxonomy of current explanation approaches in cybersecurity is shown in Figure 4. We analyse each category of ante hoc and post hoc explanations. The two categories have different classification criteria, based on the structural characteristics of models³³ and the target models.³⁴

1. *Ante hoc explanation type*: The ante hoc explanation models are algorithmic transparent with human-interpretable mechanisms.³⁵ The decomposability of an ante hoc explanation model requires all parts of the model, including the model structure, model parameters, every input to the model, and every dimension feature allowing intuitive explanation.
2. *Post hoc explanation type*: In the applications of ML in cybersecurity, architectures of RNNs and convolutional neural networks (CNNs) are prevalent.³⁶ These architectures of neural networks have a common deficiency: The neural networks are opaque, making them hard to interpret for practitioners. These predeveloped models are target the model that needs to be described by explanations. Other ML model or mixed neural network is the third type for target models.

CNNs: In CNN, each neuron in a layer receives input from a restricted subsection of the previous layer.³⁷ Layers with neighbourhoods in the inputs data form a strong spatial structure. In cybersecurity, CNN models can be used for malware detection from raw Dalvik bytecode of Android applications.³

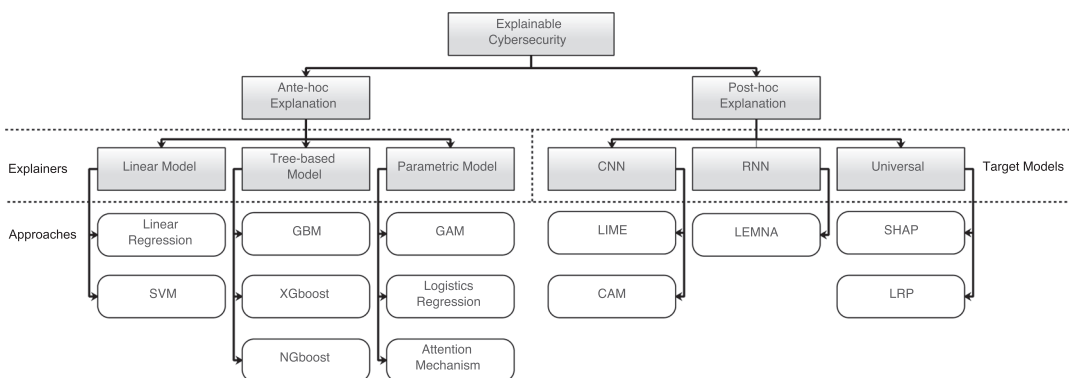


FIGURE 4 Taxonomy of explainable approaches in cybersecurity: ante hoc explanations are classified by the type of explainers models; post hoc explanations are classified by the type of the target model. CAM, class activation mapping; CNN, convolutional neural network; GBM, gradient boosting machine; LEMNA, local explanation method using nonlinear approximation; LIME, local interpretable model-agnostic explanation; LRP, layerwise relevance propagation; NGBoost, natural gradient boosting; RNN, recurrent neural network; SHAP, SHapley Additive exPlanation; SVM, support vector machine; XGBoost, extreme gradient boosting.

RNNs: The network with loops in RNNs connects neurons from the output layer to the input layer. RNNs with a recurrent structure mainly operate sequential data³⁸ and process information across predictions. RNN models have been used for security applications of sequential data (e.g., the discovery of vulnerabilities in software).⁶

Universal: Some approaches process mixed Neural Networks rather than typical CNNs or RNNs. The universal methods are not limited to the type of target models. This category is universal to CNN, RNN, mixed neural networks and ML models.

The details of each explainable approach, including security applications, and pros and cons, of each explainable approach will be introduced in the Sections 3 and 4, respectively.

2.4 | Properties of explanations in cybersecurity

We look at the explanations' properties. The properties can be used to identify the suitability of an explanation method or to validate how good it is.^{39,40} The explanations for predictions usually correlate the feature contributions of a single instance to its classification in a form that human can easily understand. Besides, the explanations include a set of data samples. We can detect malware using support vector machine (SVM) models and explain malware using local proxy methods. Alternatively, we can use models equipped with an explanation module (e.g., a gradient boosting model). The properties of explainable approaches are as follows:

1. **Global explanation:** Global techniques allow us to better understand the entire model, for example, visualisations of the distribution of weights in neural networks and neural network layers propagation.
2. **Local explanation:** Local techniques focus on a smaller group of data that is usually overlooked by global techniques, for example, looking at a single instance and providing explanations.
3. **Model-agnostic:** Model-agnostic explanations can be applied in any trained model. They usually examine input features and output distribution of the data. Recent techniques for model-agnostic explanations are about the black box that needs to be interpreted.
4. **Model-specific:** These techniques are only suitable for specific class of models. They generally focus on algorithm or intermediate structure of the model. The explanations with intrinsically interpretable algorithms are model-specific.
5. **Security:** Explainers threaten the security of their target systems.⁴¹ Due to increasing attention to the explanation of Deep Neural Network (DNN) model, the key factors to the classification model's decision can be identified, which can easily construct malicious adversarial samples to attack the classifier of the security system. Security issues, such as cyber attacks in Cyber-Physical Systems (CPSs),⁴² pose a huge threat to the systems.
6. **Transparency:** Explainable technologies are used to solve the trust problem for predictions with transparent methods. They help models to become more transparent. We usually use explainable approaches to solve the problem of transparent box design.

Sometimes, ML models have other important properties, such as robustness, reliability, causality, generality, and scalability. Models could maintain certain levels of performance, for example, some models are robust and reliable to small changes in the input data or parameters, some models control the classification changes due to perturbations affecting the model

predictions (causality). These properties vary when dealing with data sets or models^{30,43} and are not general enough, therefore, they will not be discussed in this survey.

3 | ANTE HOC EXPLANATION APPROACHES

Ante hoc explanation deals with the inner working of models to interpret its results. It mainly refers to whether the model outputs are interpretable. Developing an explainer means providing users with potential recommendations and even interventions,⁴⁴ which improve their situation. Meanwhile, explainable ML systems provide comprehensive understanding. Users are able to understand clearly how model outputs get translated into actions,⁴⁵ and how to obtain the outcomes generated by the system. It also enables users to understand and know why a decision is made.

These explainers are more interpretable than other ML models.²⁰ For example, it is easier to understand the output of a linear model than the output of a neural network model. However, since a mathematically straightforward model is more interpretable,⁴⁶ these ante hoc explanation models require a more explainable feature engineering (better understanding features) than others.⁴⁷ Basically, before the model is run, human decisions derived from ante hoc explanation come into play.

We divide these explainers into three categories: linear model, tree-based model, and parametric model.

3.1 | Linear model

3.1.1 | Linear regression

Security application: Android malware detection and vulnerability discovery

Linear regression models predict the instance as weighted feature inputs.⁴⁸ The linearity relationship makes the explanation of learning models much easier. Computer experts have long used linear regression algorithms to solve quantitative issues.

Method: Linear models can establish the dependence relation of target models y on the feature set x , which is a linear relationship. The function of a linear model is as follows:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \varepsilon. \quad (1)$$

The predicted result of an instance is the weighted features, wherein β_i represents the learnt weights of features, the first weight β_0 is not multiplied with any feature and can be seen as an intercept, and ε , which follows a Gaussian distribution, is the difference between predictions and real outcomes, like, an error.

Advantages: The linear model makes decision-making more intuitive for humans as the model has weighted features. They are transparent to estimate the feature weights and can get optimal weights theoretically. Plenty of applications have used linear regression models, and it is shown that linear regression models are accepted for prediction in a lot of scenarios. There are collective experience and expertise at high level of expertise. It has been implemented in software. In addition linear regression methods can be found in various programming languages, such as Python, R, Java, JavaScript, and so forth.

Disadvantages: Since linear regression only represents linear relationships for the features of input, applications with interaction and nonlinear relationships cannot use it. Besides, the predictive performance of linear models is often suboptimal because the learned relationships are restricted and often oversimplify the reality.⁴⁹

3.1.2 | SVM (Support Vector Machine)

Security application: Android malware detection, intrusion detection and binary analysis

SVMs utilise the so-called support vectors from the training samples to look for the decision boundary.⁴³ It finds a hyperplane with a maximum margin around the SVM decision boundary with a set of different kernels.⁵⁰

Method: SVM in security only use its one method, linear SVM. We get a training data set with n data points. The hyperplane of SVM can be regarded as a set of data points $\vec{x}$, which satisfy:

$$\vec{w} \cdot \vec{x} = -b, \quad (2)$$

wherein $\vec{w}$ is the (not necessarily normalised) normal vector of the decision hyperplane and b is an intercept term. w and b pair to the training points x .

The application of SVM in Drebin is shown in Figure 5. By virtue of the simple detection function of a linear SVM, it determines the contribution of each feature.² The linear SVM provides an explanation of detection from the perspective of malware families.

Advantages: SVM is a linear margin-maximising classifier. It works well when there are clear margins of separated classes in various real-world problems. SVM application can tremendously reduce the need for training instances with labelled data in the standard inductive settings.⁵¹ It is easy to know the importance of each feature from linear SVM, which is what convinces the users most because it is the model's own explanation.

Disadvantages: In security applications, classifiers require high accuracy, but linear SVM is not as accurate as nonlinear SVM. DREBIN is an efficient and explainable malware detection using SVM,² and its successful rate reaches 97%. But SVM is not suitable for large data sets.

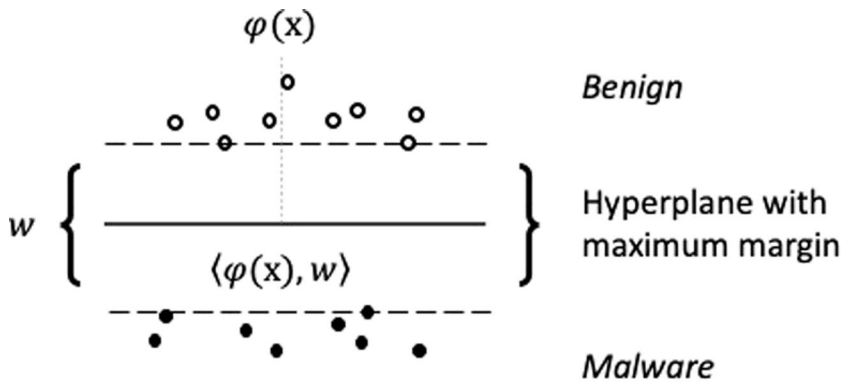


FIGURE 5 A separating hyperplane with maximum margin (SVM) on the malware detection system. Constructing an input x as a vector $\phi(x)$. Mapping the input to a vector space to see its classification (benign or malware) from the hyperplane. SVM, support vector machine.

Besides, if the number of features for each instance exceeds the number of training samples, the performance of the SVM algorithm will decrease.⁵²

3.2 | Tree-based model

3.2.1 | GBM (Gradient boosting machine)

Security application: Android malware detection and vulnerability discovery

GBM is a general tree-based method working on boosting technique. Each new tree in GBM can be used to modify the original data set.⁵³ GBM trains the model in a gradual and additive manner.

Method: In gradient boosting, the loss function can be minimised by gradient descent, which indicates how good the coefficients of the model reflect the data. What is optimised determines the information of explanations. The design idea is to iteratively build the model in sequence, with each updated model reducing errors based on the previous model until the loss is negligible.⁵⁴

Advantages: GBM is a method that is easy to understand, and the prediction capability is efficient through the use of its clone methods. Besides, GBM makes its explanations on predictions easy to control as well.

Disadvantages: Gradient boosting models are continuously improved to minimise all errors, which can lead to overfitting by focusing too much on outliers. In addition, GBM has to perform grid search and can only test limited values.⁵⁵

3.2.2 | XGBoost (Extreme gradient boosting)

Security application: Android malware detection and vulnerability discovery

XGBoost is one of the tree-based methods and is known as the regularised version of GBM.⁵⁶ XGBoost is an efficient and easy-to-use ML algorithm. Compared with other algorithms, it can achieve high performance and accuracy.

Method: In order to train the model, it is necessary to define the objective function to measure how well the model fits the training data. Objective functions in XGBoost contain two parts: 1) training loss and 2) regularisation. One commonly used loss function is the logistic loss. For the regularisation, it controls the complexity of the model, so it can help us avoid overfitting.

Advantages: XGBoost prevents this model from overfitting with built-in Lasso Regression (L1) and Ridge Regression (L2) regularisation. Users are allowed to run cross-validation at each iteration in the boosting process.⁵⁷ In this way, it is easy to get the exact optimal boosting iterations number. Besides, it is much faster than GBM because XGBoost utilises multiple central processing unit (CPU) cores when executing a model. In addition, XGBoost supports the following main interfaces: CLI, C++, Python, Julia, R, Java, and so forth.

Disadvantages: Although XGBoost is fast to some extent, it does not perform well in large data sets as it is time-consuming for training. Once XGBoost fails, it takes quite a while to recover. What is more, people have to manually create label encoding for categorical features before feeding them into the models.

3.2.3 | NGBoost (Natural gradient boosting)

Security application: Android malware detection and vulnerability discovery

NGBoost is a general probabilistic prediction algorithm with explainability.⁵⁸ It allow the prediction of uncertainty estimates, rather than point estimates of typical regression models.

Method: Targeting at the parameters of the conditional distribution, NGBoost generalises gradient boosting on probabilistic regression. The algorithm has three modular components: (1) base learner, (2) parametric probability distribution and (3) appropriate scoring rules.⁵⁸ Predictions are made on new inputs form a conditional distribution. The parameters of a conditional probability distribution are generated by the additive combination of the outputs of multiple base learners. This algorithm is an explainable method for probabilistic predictions.

Advantages: NGBoost can work with any base learner and any scoring rule, and can perform better than existing methods (e.g., elastic net) for probabilistic prediction on point estimation tasks. It can also solve the joint prediction problems.

Disadvantages: There is a technical problem with NGBoost, That is the natural gradient loses its invariance property with a finite step size.⁵⁸ In addition, NGBoost has a limitation that it can only predict conditional distributions by generalising gradient boosting.

3.3 | Parametric model

3.3.1 | GAM (Generative additive model)

Security application: Android malware detection

GAMs are extensions of the linear model. Linear models mean that increasing the value of an instance always has the same effect on their predictions, no matter what the value is. However, the values of the predictor variables being predicted in GAMs are given by the accumulation of many functions designed for predictors, so class activation mappings (CAMs) can be considered decomposable models.⁴⁹ The relationship between features and predicted outcomes of GAMs is usually nonlinear and the features interact one another.

Method: GA relax the restriction that features can only be simply weighted. By building a sum of each feature's function, They can predict and explain.

Advantages: The extensions of linear models themselves are general. For almost all cases of linear models, there are extensions that fit them. The linear model and its extensions have been established in many research areas. We can apply this model for inference, but not for making decisions. We get confidence intervals for weights, that are prediction intervals.⁵⁹

Disadvantages: Most modifications to linear models make models less interpretable. Besides, GAMs rely on assumptions on data-generating processes, therefore, when assumptions are violated, explanations for predictions are no longer valid.

3.3.2 | Logistics regression

Security application: Android malware detection

Logistics regression is a model that uses a logistic function. It provides two possible outcomes for classification problems. Logistic regression can be regarded as an extension of the linear regression model in classification.

Method: Logistic regression does not fit any hyperplanes or lines.⁶⁰ Instead, it uses a logistic function to obtain predicted values for outcomes between 0 and 1. The function of logistic regression can be written as follows:

$$\text{logistic}(\eta) = \frac{1}{1 + \exp(-\eta)}. \quad (3)$$

Logistic regression models find the right decision boundary between two sides. The interpretation of the weights in logistic regressions is different from that in linear regressions because the result outputs of logistic regressions are between 0 and 1. Weights no longer affect probability linearly, and are converted to probabilities according to their logistic functions.

Advantages: Logistic regressions have been widely adopted by many people. Firstly, the merit of linear regression models also applies to logistic regression models. Secondly, logistic regression models are used for classification and can provide probabilities as well.⁴⁵ Compared to machine learning models that only provides final classification results, this is an additional advantage. For example, the probability of being 99% and just above 50% respectively for an instance being classified in one class is very different.

Disadvantages: Logistic regression models are difficult to interpret because weights are explained by multiplication rather than addition. Logistic regression can be affected by complete separation. The logistic regression models are not trained any longer if there is a feature that separates two classes, because it could result in an infinite weight for an optimal solution. If such a feature exists and is useful, it would be really a drawback. But if users (e.g., the developer and the analyst) had a simple rule to distinguish between the two classes, it is not necessary for them to use ML techniques.

3.3.3 | Attention mechanism

Security application: Android malware detection

Attention mechanism is a useful tool and a self-explanatory method in a neural network.⁶¹⁻⁶³ Due to the complexity and low transparency of neural network model structure, one of the effective method to bring a model's own explanation is to introduce the mechanism of attention⁶⁴ as an additional interpretative module. It was first introduced by Bahdanau et al.⁶¹ When processing data with deep learning, the attention mechanism is used to pay more attention to certain factors.⁶⁵ Attention mechanism is widely used in diverse fields, especially in NLP systems.

Method: The attention mechanism allocates limited information-processing resources to more important tasks by determining which part of the input is more important. Hence, it solves such problems as not being able to remember long sentences. With the attention mechanism, attention is integrated into the encoder-decoder as a converter. It illustrates how relevant is the input elements to the output.¹⁸ By utilising the attention vector, it estimates how related an element is to other elements. The attention mechanism takes the weighted sum of element values by the attention vector as the approximation of the target as well.

Advantages: The attention mechanism shows excellent classification results and explanations.⁶⁶ It identifies the inputs that are most relevant to the task, thereby improving performance in processing.¹⁸ It extracts more critical and important information by assigning different weights to each part of the input. Besides, it can make the prediction of the model more accurate, making the prediction easier to understand. Besides, it is a very popular explanatory tool, for example, providing an interpretable recommendation system by training picture descriptions, and applying attention mechanisms to picture looking and speaking,^{67,68} to improve the history setting based on the recommendation system.⁶⁹

Disadvantages: Attention mechanism has been widely applied in many fields, such as NLP, computer vision, recommendation system and so forth.^{70,71} One of the early applications in computer security was from Wu et al. 's work¹⁸ which designed a new explanatory model, XMAL model, to explain Android malware in conjunction with the attention mechanism. However, whether the attention mechanism itself is explanatory is indeed a topic of discussion in recent years.^{72,73} Wiegrefe and Pinter⁷³ claim that random distributions or counter-perturbation distributions indicate that attention can be an "explanation."

4 | POST HOC EXPLANATION APPROACHES

Post hoc explanation model is not interpretative and needs a post hoc interpretability method. Post hoc techniques allow models to be trained. For example, LIME (a local interpretable model-agnostic explanation is a landmark post hoc technique, which works by perturbing a single prediction feature to determine the most contributed features to the model's decision. Post hoc explanation tends to be more flexible than the ante hoc one⁷⁴ because it can be applied to various kinds of models. Meanwhile, it also provides less explanation of the model itself. Among the growing number of approaches proposed to generate explanations for various ML classifiers, post hoc explanation approaches⁷⁵ have been the growing subject of recent research.⁷⁶ By generating a well-trained explanation of the predictive model, rather than treating it as a black box, these systems can be used in any situation because they are inherently flexible enough.⁴⁹

Post hoc explanation provides surrogate models to the target models. By training a surrogate model, it is possible to obtain predictions that approximate the black-box model.⁷⁷ A DNN can handle complex nonlinear relationship by modelling multiple hidden layers,⁴⁵ while it cannot be explained. A largely used DNN model is the RNN. Moreover, some approaches to processing CNNs are typically used in image analysis.⁸ The remaining is not restricted to the type of target models, as it is universal to CNN, RNN and ML model.

4.1 | CNN target model

Among many malware detection systems, one of the static detection methods is image analysis detection, for example, Microsoft malware detection⁷⁸ and Android malware detection.⁷⁹⁻⁸¹ Malware can be visualised as a grayscale image based on its executable binaries.⁸² According to the malware's characteristics and behaviour, image processing systems with a CNN architecture can effectively identify the image and classify the malware into different families.⁸³ Interpretive methods targeting CNN models include LIME and CAM, which can provide suitable explanations for the CNN detection systems.

4.1.1 | LIME

Security application: Android malware detection, PDF malware detection and vulnerability discovery

LIME is a well-known interpretable model. It is used to explain black-box ML models for a single instance,⁸⁴ and it focuses on training local models to explain how individual instance makes predictions.

Method: LIME is able to give model-agnostic explanations. Suppose g is the explainable model for an instance x . The model complexity $\Omega(g)$ is usually very low.⁸⁵ There is a set of potential explanation G , for example, linear models. The neighbours around an instance x can be defined by the proximity π_x . The local interpretable surrogate models for a single instance is as follows:

$$\text{explanation}(x) = \arg \min_{g \in G} L(f, g, \pi_x) + \Omega(g). \quad (4)$$

Minimising the loss L helps measure how approximate the prediction of the target model f is to the explanation.

Advantages: LIME benefits from the experience of training explainable models. The fidelity measure on explanation helps us get a good sense of the reliability of interpretable models in explaining black-box predictions, and it can be applied to explaining any black-box model.

Disadvantages: First, it is obvious that LIME does not suitable for providing a global explanation. Second, LIME only optimises the loss part in practice.¹⁴ The user must therefore determine the complexity, for example, by limiting the number of features. In addition, data sampling follows a Gaussian distribution from LIME, which ignores the correlation or dependency among the features, and can lead to unexpected data points between samples when learning local explanation models.

4.1.2 | CAM

Security application: Android malware detection

CAM is a popular saliency method that uses global average pooling in CNNs to provide explainable information. Class activation maps for a particular category represent areas of the image that CNN uses to identify the category. Discriminative features for categories can be identified because of the simple connectivity structure.⁸⁶

Method: The global average pool in CAM outputs feature maps for each unit of the last convolutional layer.⁸⁶ The explainable output is produced by the weighted sum of feature values. The class activation map is obtained by calculating the weighted sum of the feature maps of the last convolutional layer. The CAM model is expressed as follows:

$$M_c(x, y) = \sum_k w_k^c f_k(x, y). \quad (5)$$

This is an explanation for an instance x at pixel (x, y) , which generates heatmaps by the pixelwise activation $f_k(x, y)$, wherein k is a unit in the last convolutional layer that acrosses spatial coordinates (x, y) . w is the weighting coefficient on the unit k for class c .

Advantages: It incorporates network activation into visualisation. The features are fed into the completely connected neural network layer handled by the softmax activation function.⁸⁷ The importance of the weights can be found by projecting back the weights of the output layer onto the feature map of the last convolution layer. Moreover, the idea generated by CAM is simple, and it has been applied to most recent CNN architectures since they use a single full connectivity layer.

Disadvantages: CAM models lack full understanding of their features as they are expressed as part of the images. Currently, CAM is rarely used in cybersecurity because security systems, such as detection systems, are rarely studied based on image classification.

4.2 | RNN target model

4.2.1 | Local explanation method using nonlinear approximation (LEMNA)

Security application: PDF malware detection and binary analysis

To account for the fidelity, LEMNA¹⁴ approximates each input instance of the local decision boundary by training a mixed regression model. Fused Lasso is introduced in LEMNA to effectively make up for the deficiency of LIME, which is used to deal with RNN model features. LEMNA is able to generate translatable information (i.e., the intermediate transformation of features) to interpret input instance predictions.

Method: LEMNA approximates local decision boundaries by combining multiple simple interpretable models (i.e., a mixture linear regression model with a weighted sum):

$$f(x) = \sum_{j=1}^{\pi_j(\beta_j \cdot x + \epsilon_j)} \quad (6)$$

β is the regression coefficient of the training samples x . There are linear approximations of the local boundary combined in a mixed linear model. The weight of each corresponding model is π_j . LEMNA illustrates a high correlation in most of the features by experiments (e.g., VulDeePecker¹¹).

Advantages: Through LEMNA, security analysts can understand DNNs and even patch the errors from security systems.⁸⁸ The constrained nature of Fused Lasso makes the returned features highly relevant.¹¹ It can be applied to nondiscrete feature models.

Disadvantages: Although LEMNA focuses on the “principal reason” in predictions of the input, the given features require further analysis of the key features so that the user can understand the specific reasons. This approach relies heavily on the understanding of the features themselves. LEMNA providing features with high relevance may help attackers find weaknesses in classification models, thereby weakening defenses. Besides, compared with other explainable methods, implementing LEMNA is time-consuming, and usually takes around 10 times longer than executing other models (e.g., the average runtime of four security systems data sets Drebin+, DAMD, VulDeePecker and Mimicus+ are 180 s on LEMNA and 4.8 s on SHapley Additive exPlanation [SHAP]).³⁰

4.3 | Universal target model

4.3.1 | SHAP

Security application: Android malware detection and vulnerability discovery

SHAP is a method to explain individual instance predictions, and generates optimal Shapley values from the game theory.⁸⁹ Shapley values provide users with a fair distribution of the features. Specifically, shapley values identify which features determined the output of a model's prediction, and whether the impact of features is positive or negative.⁹⁰

Method: SHAP is represented as a linear additive feature attribution method. By simulating whether the features values are present or not, this method could compute Shapley values. The main contribution for predictions are the key features with a large Shapley value. That is the core idea of feature importance in SHAP. By running SHAP on the input data, we get a matrix of Shapley values with instances as rows and features as columns.⁹¹ We can explain a model by analysing the Shapley values in this matrix.

Advantages: SHAP generated in game theory has a solid theoretical foundation, so it can explain predictions for any ML model. Shapley values for a single instance can provide explanations for a prediction, and can be assembled into global explanations for a full data set. Besides, SHAP has a fast implementation of explanations by comparing an individual's prediction with others' predictions,⁸⁸ which makes it easy to get Shapley values for the model's global explanations, e.g., feature contributions, feature dependence and dependence plots. Furthermore, the global explanation in SHAP is consistent with the local explanation of the same data set.

Disadvantages: Unfortunately, Shapley values can be misinterpreted for new data, if different explainers are used. There are three main explainers of SHAP: KernelSHAP, TreeSHAP and DeepSHAP. KernelSHAP explainer is slow and ignores feature dependence,⁴³ which makes KernelSHAP dissatisfactory to use when users want to calculate Shapley values for some data instances. All global explanations in SHAP (e.g., feature importance) need to compute Shapley values for a large sample. TreeSHAP relies on conditional expectation predictions to change the value function. With the changes for functions, features can get a TreeSHAP value different from zero without the influence on the predictions. TreeSHAP explainer can produce unintuitive feature attributions. Third, Deep SHAP is only an approximate algorithm to compute SHAP values for deep learning models.⁴³

4.3.2 | LRP (Layerwise relevance propagation)

Security application: Android malware detection and vulnerability discovery

LRP is one of the explainable ML methods tailored to certain feature types (e.g., text or saliency maps).⁸ It works to explain the output of any neural network in its input domain.

Method: LRP is to keep the back-propagation process unchanged from the output y to the input layer R as shown in Figure 6.⁹² The magnitude of input layers is conserved at the back-propagated layers through the network and is transitive to the output layer. The LRP algorithm attributes relate to each input node by tracing back contributions to the output node of the final layer. That is, when using LRP to find out how network neurons and inputs correlate with a prediction, it can be seen as the network path to reach this neuron through the output neuron of the predicted class (Figure 6).

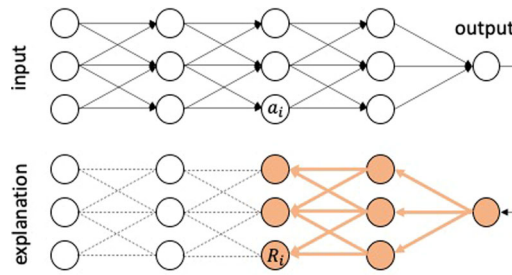


FIGURE 6 Explaining neural network predictions based on input features using the Layerwise Relevance Propagation (LRP). Apply a forward pass to a network and collect the activations for each layer. The relevance score R_i is obtained at the output by back-propagating the network using the conserved propagation layer. a is an activation of the corresponding neuron. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/int.23088)]

Following this basic LRP rule, the sum of relevance R across layers is conserved. Its contribution to neuron j as follows:

$$R_j = \sum_{k=1}^K \frac{a_j w_j k_j}{\sum_0 a_j w_j k_j} R_k, \quad (7)$$

where k is a neuron of any consecutive layers, and w denotes weights between the two neurons in adjacent layers. We can calculate relevance R for each neuron in previous layer by using the formula iteratively.

Advantages: LRP provides explanations and extensions for highly complicated neural networks.^{92,93} It accurately and quantitatively informs features that contribute or counteract the model in making decisions.⁹⁴ The target model's network structure is not strictly limited. LRP works perfectly for CNNs as well as for RNNs.⁸⁸

Disadvantages: LRP is mainly applied in the area of the image. Nowadays, it has expanded to machine translation and NLP, but it is rarely used for security systems and does not perform as well as other interpretable methods.

5 | COMPARISON OF EXPLAINABLE APPROACHES

5.1 | General explanation evaluation

Various explainable approaches have been proposed for explanation in cybersecurity. To help researchers select proper explanatory tools, we summarise the performance of the aforementioned approaches with their properties, as shown in Table 1.

There is a fundamental difference between ante hoc and post hoc explainable approaches: post hoc explanations are model-agnostic that they assume access to querying the target model on any input (e.g., LIME and SHAP), while ante hoc explanations are model-specific because the explanation generator and the predictor are the same model with varying degrees of transparency.

TABLE 1 Summary of explainable approaches with seven properties: global explanations, local explanation, model-agnostic, model-specific, traditional machine learning, security and transparency

Approaches	Global	Local	MA	MS	TML	Security	Transparency
Linear regression	◦	•	◦	•	✓	–	★★★★★
SVM	•	◦	◦	•	✓	–	★★★★★
GBM	◦	•	◦	•	✓	–	★★★★★
XGBoost	◦	•	◦	•	✓	–	★★★★★
NGBoost	◦	•	◦	•	×	–	★★★★★
GAM	◦	•	◦	•	×	–	★★★★★
Logistics regression	•	◦	◦	•	✓	–	★★★★★
Attention mechanism	•	◦	◦	•	✓	–	★★★★★
LIME	◦	•	•	◦	×	×	–
CAM	◦	•	•	◦	×	×	–
LEMNA	•	◦	•	◦	×	×	–
SHAP	•	◦	•	◦	×	×	–
LRP	◦	•	•	◦	×	×	–

Note: MA, model-agnostic; MS, model-specific; TML, traditional machine learning; ◦, applicable; •, not applicable; ✓, positive; ×, negative; a five-star rating system = level of transparency; –, none.

Abbreviations: CAM, class activation mapping; GAM, generative additive model; GBM, gradient boosting machine; LEMNA, local explanation method using nonlinear approximation; LIME, local interpretable model-agnostic explanation; LRP, layerwise relevance propagation; NGBoost, natural gradient boosting; SHAP, SHapley Additive exPlanation; SVM, support vector machine; XGBoost, extreme gradient boosting.

In general, the ante hoc explanation approaches incorporate an explanation generation module into their architecture so that they can provide explanations for their predictions.

5.2 | Post hoc explanation evaluation criteria

The post hoc explanation methods have their unique evaluation criteria, as they tie in with the model after training (target models).²⁰ Explainers processing mechanisms are complicated due to different explainers of one classifier with different processing mechanisms. Their explainable technologies address the common issues on the view of security systems. Users with explainable technologies can not only understand the mechanisms of systems but also make better decisions. The central part of the explainable technologies is to train explainable models for the security system. This study acquires existing explainable methods and creates an advanced explainable method as well. Therefore, we illustrate the evaluation of each explainer implementation on a security system with evaluated properties. We report six metrics that demonstrates desirable aspects of the post hoc explanation: accuracy, sparsity, completeness, stability, efficiency, and fidelity.

1. *Sparsity*: It is used to measure the local explanation. Explanations with sparse are especially helpful for learning large graphs, since manually checking all nodes is not feasible.⁹⁵ Sepcifically, this measure was defined by subtracting the amount of objects identified in

the heat-maps explanations $m_0 \vee m_1$, divided by the total number of nodes in the graph $|V|$, that is $1 - \frac{m_0 \vee m_1}{|V|}$.

2. *Completeness*: In security systems, when an explainable method is able to generate nondegenerated explanations for all input, this explainable method is complete for a predicted model. If there is an input that cannot be processed by the explainable method, an adversary (e.g., with random perturbations) may trick the method to be unable to derive a valid estimate, resulting in degenerated results.¹¹
3. *Accuracy*: Accuracy indicates how well an explanation predicts instances. High accuracy is quite important in applying ante hoc explanations, since explanations are used (instead of ML models) to make predictions.⁹⁶ Low accuracy may occur when: (1) the model's prediction accuracy is also low; or (2) a black-box model is interpreted. This case needs to consider fidelity which will be discussed later. Accuracy is the most important criterion for acquiring reliable explanations that accurately capture features relevant to predictions.
4. *Stability*: Small changes in input samples sometimes can strongly alter predictions. High stable explanations mean slight changes in an instance features do not significantly change the explanation. Some factors can cause low stability, for example, uncertainty in the explainable method components in data sampling steps.⁹⁷ Explainable explanations always require high stability.
5. *Efficiency*: Explanations need to be available in reasonable time when operating an explainable model. The time differences between different explainable modes are significant. If a method provides an explanation without delaying a workflow, we consider that method efficient since there are usually no strict requirements on runtime.
6. *Fidelity*: Fidelity shows how approximate the explanation is to a black-box model. High fidelity makes us more confident in the explainable model. If a predictive model has high accuracy, the explanation with high fidelity also has high accuracy. Some explanations provide local fidelity for model predictions, which means they only explain a small set of data predictions or a single instance prediction.

We examine the quality and properties of each explainable method by looking at these metrics. An experiment result for certain security systems (Drebin, Mimicus, VulDeePecker and DAMD) with post hoc explainers by Warnecke et al.¹¹ is shown in Table 2. The relevant values of the table are the average quality of these methods for the security system

TABLE 2 Evaluation on post hoc approaches¹¹

Approaches	Sparsity	Completeness	Accuracy	Stability	Efficiency (s)	Fidelity
LIME	0.772	–	0.582	0.353	2.1×10^{-1}	Middle
CAM	–	–	–	–	–	High
LEMNA	0.612	–	0.702	0.289	1.8×10^2	Middle
SHAP	0.757	–	0.823	0.279	4.8	Middle
LRP	0.873	✓	0.454	1	5×10^{-2}	Middle

Note: The values for accuracy, sparsity and stability range from 0 to 1. The levels for fidelity are low, middle and high. Abbreviations: CAM, class activation mapping; LEMNA, local explanation method using nonlinear approximation; LIME, local interpretable model-agnostic explanation; LRP, layerwise relevance propagation; SHAP, SHapley Additive exPlanation.

data sets. The efficiency shows each sample's average run-time in seconds, ranging from microseconds to minutes. On the basis of the efficiency of this table, LRP is the most efficient one with computing results in less than 50 ms. From the accuracy, the scores for LEMNA and SHAP are 0.702 and 0.823, much higher than the others. All these approaches perform similarly in sparsity, mostly around 0.7. From these evaluation metrics, we conclude that: (1) in the test data set, SHAP is sparser than LEMNA, but less stable; (2) LRP is most efficient but has low accuracy; (3) CAM has the highest fidelity, but only suitable for CNN-based target models; (4) LEMNA is inefficient and has low sparsity. All these methods are applicable to tabular data (i.e, structured data in tables) except CAM for image data.

6 | TIMELINE AND OPEN ISSUES

The following section 6.1 explores in more detail the issues in the fundamental causes of the prediction–explanation gap. We review the research development and all the important events of explanatory technology in security applications with a clear timeline. Then we discuss open issues about the explanation in cybersecurity.

6.1 | Timeline of explanation in cybersecurity

Explanatory methods were developed as a result of the need to understand and trust ML techniques. For many decades, basic explanation methods have been proposed, such as logical regression. Since 2014, interpretative methods in the field of computer security have been explored and studied. Since the explainable artificial intelligence was proposed as a concept by the Defense Advanced Research Projects Agency,⁹⁸ the research field of explanation technology has entered the stages of development. Figure 7 is the entire explainable timeline for cybersecurity development.

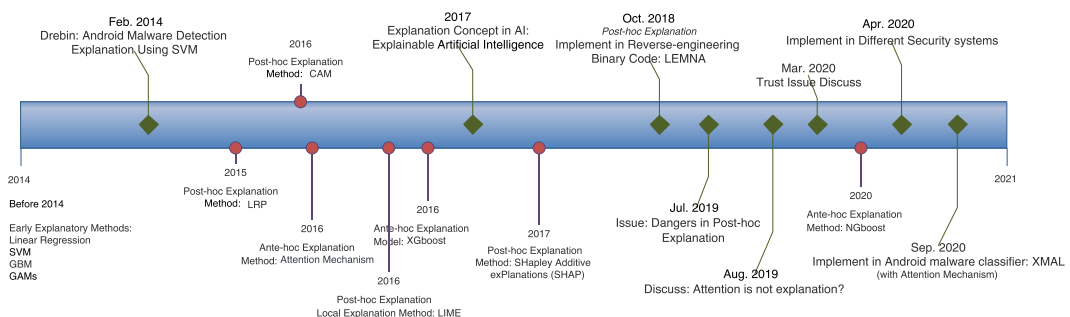


FIGURE 7 Timeline of explanation in cybersecurity. Point of red circle means designing an explainable method (marked with ante hoc explanation or post hoc explanation) and point of green rhombus means events about experiments or discussions on explanatory methods. AI, artificial intelligence; CAM, class activation mapping; GAM, generative additive model; GBM, gradient boosting machine; LEMNA, local explanation method using nonlinear approximation; LRP, layerwise relevance propagation; NGBoost, natural gradient boosting; SVM, support vector machine; XGBoost, extreme gradient boosting. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

SVMs, linear regression, GAMs and GBMs are examples of traditional explainable algorithms with strong theoretical properties.^{53,99,100} These models are classical and highly explanatory due to their high transparency and low complexity.

However, those traditional ML models do not add trust in their predictions. That is an important problem in a decision-making model in security applications. As a result, many new explanatory methods are proposed to solve this problem. Since 2014, people have applied these methods to security systems, with more experiments that they try to understand input features and prediction of complex models.

The most typical experiment is the tool Drebin², designed for detecting Android malware in 2014. It utilised the binary classifier of linear SVM to detect Android applications and determines the feature contributions to the classification outputs through the coefficient of the detection function, and to explain the classification of Android malware. In Ribeiro et al.,⁸⁴ a model-agnostic explanation was proposed, Known LIME, the linear model coefficient directly indicates the importance of a given feature. Any DNN can be explained using this black-box method. Diverse areas have benefited from its use.

Subsequently, Lundberg and Lee¹⁰¹ further introduced another interpretable approach to the black-box model in 2017, that is SHAP. SHAP is designed to combine LIME and Shapley value, a concept from the game story, and can be interpreted as its contribution to prediction.¹⁰²

After the concept of Explainable Artificial Intelligence proposed in 2017, more advanced approaches to security systems are generated. LEMNA,¹⁴ an interpretable model for security systems, was proposed in 2018. It uses a mixture regression model for approximating decision boundary. It also brings a deep discussion about evaluations of explanation methods in cybersecurity.

Several issues have been explored and discussed when we have implemented more explanatory methods. These issues are discussed in Section 6.2.

6.2 | Open issues

Among the incremental amount of approaches proposed in cybersecurity to generate explanations for systems, some issues are exposed in cybersecurity based on previous research on explanation: (1) trade-off between accuracy and explanation; (2) trust issue; (3) attention is not an explanation; (4) not robust counterfactual explanations; (5) is the explanation an explanation of the specific features? Figure 8 demonstrates the Timeline of Open Issues of Explanation in Cybersecurity.

1. *Trade-off between accuracy and explanation*: When using ML models to make predictions, the equilibrium between accuracy and explainability is an unbalanced. Simpler predictive algorithms are more easily interpreted by human beings, while they are prone to be less accurate than the complex and advanced methods that are hard to explain.¹⁰³ The explanation is an assessment of the model, depending on how easily the model is explained to humans. Model accuracy is a quantitative measure¹⁰⁴ that explains the probability of a model predicting a correct outcome.⁹⁵ Figure 9 shows the trade-off between the explainability of the model's predictions and its accuracy. Some ML models usually perform better than others, while they also have different explainability.

Our goal is to explain. We tend to choose more rigorous and simpler statistical learnt models, since they are more easily interpreted by humans. Classification requires to choose more flexible and complex statistical learning models. So we will

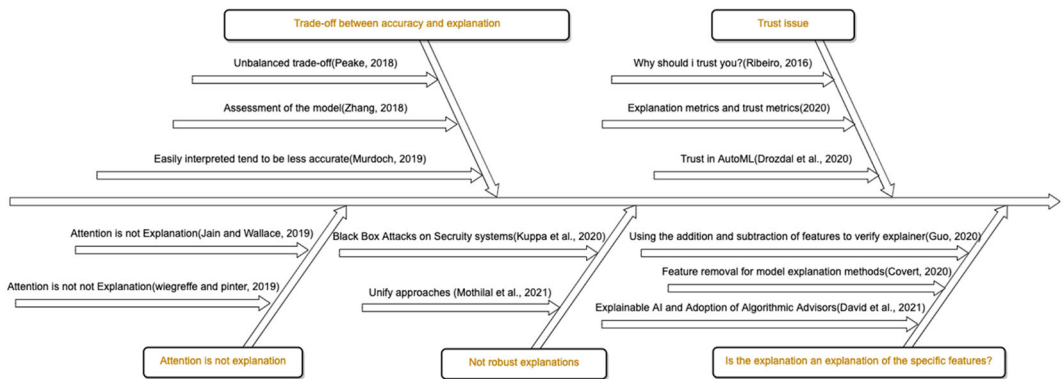


FIGURE 8 Open issues of explanation in cybersecurity. Each issue follows a timeline with two or more questions being collected in different paperworks [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jm.23088)]

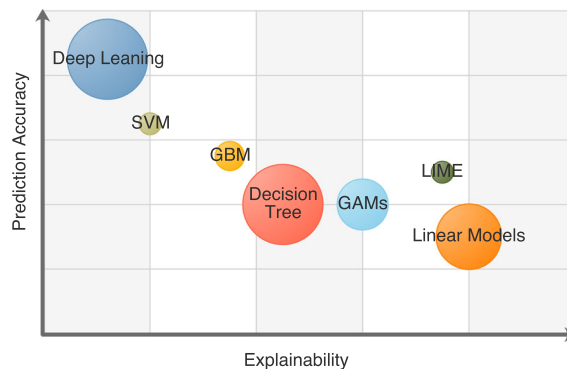


FIGURE 9 Trade-off between the explainability of the model's predictions and its accuracy: high-accuracy models usually have low explainability. GAM, generative additive model; GBM, gradient boosting machine; LIME, local interpretable model-agnostic explanation; SVM, support vector machine. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jm.23088)].

consider simple and easy-to-understand ML methods to explain more complex models. For SVM, it has a significant effect on binary classification.¹⁰⁵ However, its explainability is weak. Therefore, we adopt a linear SVM method to optimise its structure with high prediction accuracy and better explanations. For example, in Drebin, the Android malware detection tool, the experiment results were significant (94%² or 97%¹⁰⁶). Linear SVM was not as good as nonlinear SVM in accuracy in the past 2 years. Some new models have been designed for Android malware detection. For accuracy, it does not mean that a model performs well in training, but more importantly measures how well it performs in evaluation. In reality, we often want to obtain accurate prediction results through explanatory models. However, it is hard to reach high accuracy and high explanation. When performance and complexity coexist, it seems inevitable so far that explanations encounter a downward trend.

2. **Trust issue:** Before LIME was proposed, there were many studies exploring trust in models. In considering a tool for explaining black-box models, the LIME model is motivated by the

question “why should I trust you.”⁸⁴ Over the past 2 years, explanatory techniques have received increasing attention. By investigating and experimenting with users, Madumal et al.¹⁰⁷ found that a normative explanation had the following factors: understand, complete, sufficient detail and satisfying. The public trusts a system that is predictable, reliable, confident and safe.¹⁰⁸ Visualisation is also an important method of explanatory presentation.⁹⁶ In many experiments on the security system, we cannot read these features of safety systems, although they are used for training to achieve high accuracy.¹⁰⁹ Features used as explanations cannot be understood in some aspects (e.g., transforming features into grayscale images,^{109,110} binary code on reverse engineering¹¹ and sequent numbers in malware detection¹¹¹). The far-fetched use of explanatory methods is not persuasive. Therefore, in the interpretation of security system, we still need to find a better interpretation method.

3. *Attention is not explanation:* In 2019, there were two papers bring a discussion^{72,73,112,113} of the explanatory attention mechanism. Jain and Wallace⁷² proposed in “Attention Is Not Explanation” that the mechanism of Attention does not improve explanatory ability. They argue that attention weights are usually uncorrelated with gradient-based feature importance measures, and they find that very different distributions of attention can lead to the same predictions. Soon after, another group responded to this issue in “Attention Is Not Not Explanation”.⁷³ They experimentally proved that fixed distribution of attention was as good as the learnt weight of attention on some data sets, and concluded that random distribution or counterperturbation distribution did not prove that attention could not be an “explanation” in these cases. One positive attitude to the attention mechanism in cybersecurity is what Wu et al.¹⁸ applied in an experiment with an Android detection tool. The proposed explanatory method XMAL, is based on the attentional mechanism, which derives the features that have important effects on classification and the correlation between these features as well as malicious behaviours. By first selecting the feature and then training the classifier, the high detection rate (higher than Drebin) was obtained. XMAL correlates these features to explain malicious behaviour. Through observation, XMAL's prediction is highly accurate and reasonably explained through the precise extraction of a large number of feature sets, from thousands to more than one hundred. Therefore, it does not account for the mechanism of attention in terms of explanatory ability. As a result, attention may or may not be an explanation. The ability to explain depends largely on the architecture of underlying models¹¹⁴ and our understanding of the concept of explaining this.
4. *Not robust explanations:* The objective of explanation is to achieve that all features of the input and the output is a prediction of the model. There are normally multiple explanations for a given objective when it has various malicious behaviours. Different methods usually do not come to an agreement about feature importance rankings with feature-based explanations, for example, LIME¹¹⁵ discovered that the key features generated by these models were often not necessary or sufficient explanations for making decisions by changing features for generating counterfactual examples with the desired prediction. However, the issue of various feature-based explanations of a given sample can be eased by selecting the most suitable counterfactuals based on criteria for the amount of features that must be restrained, while the gap between counterfactual explanations and other feature attribution explanations needs to be alleviated.
5. *Is the explanation an explanation of the specific features?* Guo et al.¹⁴ verify the reliability of the explainer by using the addition and subtraction of features. And Covert et al.¹¹⁶ take feature reduction as a unifying principle for model explanation methods. Some ML models are easier to interpret than other models (e.g., a linear regression model vs. a neural network model).

People need to understand what information works for a model prediction. Simple models may require more work in feature engineering, while deep learning models transfer their decisions from feature engineering to model architectures.¹¹⁷

Consequently, when we say that a mathematically straightforward model is more interpretable, we are overlooking some of the human decisions that came into play before the model was even run. Sometimes, it is useful to think about target models, rather than explainable applications.

7 | FUTURE RESEARCH

Solving model transparency: Using transparent modules for local or global explanations, explainable approaches can be applied to solve the classification problem, which also solves the transparent box design problem. The explainers of the black-box models generally involve input and decision output. Although we understand the decision factors of the black-box model, we do not understand the interpreter because it is not transparent.

Explanation selection: According to the perspective of a system analyst, a good explanation strategy needs to provide reliability and maximum performance at the lowest possible cost. It is possible to incorporate multiple explanations into a decision-making process. As to choosing the interpreter, it is determined by the mechanism of the classifier, such as the type of neural network that it consists of. This process is very time-consuming. In the end, only the one with higher accuracy and fidelity can be chosen as the recommended explainer. However, this recommended explainer may not be the best choice, which is also a challenge for the future.

Formalise explanation metrics: We have considered some properties for evaluating explanations, with two metrics for two categories of explanations. But some properties are not objective, for example, completeness, which is hard to measure with. As a result, one of the challenges for future work is to formalise these metrics.

8 | CONCLUSIONS

The machine learning models for explanations are highly transparent. It is easy to understand the mechanisms of these models, which provide explanations for the complex learning model. In recent years, with the improvement of accuracy and performance requirements, more and more explainable models have been designed, especially for security systems. This paper presents these models that can be interpreted in the security domain and discusses the pros and cons for all of them. It helps different users select models for different security systems. In addition, with the deeper discussion of explanatory technology, some objective problems and future research have been proposed.

AUTHOR CONTRIBUTIONS

Feixue Yan: Conceptualisation, investigation, data curation, formal analysis and writing—original draft. **Sheng Wen:** Conceptualisation, project administration, supervision and writing—review and editing. **Surya Nepal:** Supervision and writing—review and editing. **Cecile Paris:** Supervision and writing—review and editing. **Yang Xiang:** Conceptualisation, project administration, supervision and writing—review and editing.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analysed in this study.

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